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Emerging silicon-on-nothing (SON) devices technology

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Abstract

In this paper we explain the advantages of very thin layers (in the channel and in the BOX) of the silicon-on-nothing (SON) transistors. Electrical results are also presented, with gate length down to 38 nm, with a conduction channel thickness as thin as 9 and 5 nm. It is also demonstrated that SON is better suited than bulk for accepting a metallic gate for low-voltage operation due to its intrinsic low threshold voltage. We have integrated midgap $CoSi_2$ metal gate by total gate silicidation on SON transistors with Si-conduction channel thickness down to 5 nm. Finally, we will analyse the ITRS'01 CMOS Roadmap and show that SON allows reaching the I_{on}/I_{off} specifications down to the 32 nm node and approaching closely those for the 22 nm node, that is by far impossible with bulk.
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Keywords: SON; Metal gates; Ultra-thin Si-films; Totally silicided gates; SOI; SiGe; Epitaxy; CMOS architecture; Short-channel effects; Fully depleted; BOX; Tunnel

1. Introduction

Extremely thin (nanometric) films of mono-crystalline silicon on insulator are widely recognized for their potential for end-of-roadmap CMOS transistors [1–13]. Various fabrication technologies (SIMOX, BESOI, Smart Cut, ELTRAN, ELO, etc.) have been developed permitting continuous down scaling of the Si film and the BOX, but the commercial availability of such thin films as 5–15 nm still faces problems of feasibility and dispersions. In contrast with any other SOI technology, in the silicon-on-nothing (SON) process [14], the silicon film and buried insulator, both of nanometric scale, are defined by epitaxy on a bulk substrate.

Therefore, the SON process opens access to extremely thin films (the silicon channel as well as the BOX) at the same time offering the thickness control as fine as the resolution of the epi process (less than 1 nm). In addition, not more than one (bulk type) wafer is needed to fabricate the SON layers, in contrast to two wafers in wafer-bonding type techniques. In this paper we explain why very thin layers in the SON transistor allow suppression of the short-channel effects due to the reduction of electrostatic lateral coupling (in the channel and in the BOX) and allow in this way excellent electrical performances.

Electrical results are also presented, showing functional SON transistors with gate length down to 38 nm [15], with a conduction channel thickness as thin as 9 nm. We measured in particular a large improvement in terms of short-channel effects (SCE) control due to the thin conduction channel and to the ground plane effect.

All the more, SON devices with metal gate have been integrated [16]. Extremely thin fully depleted Si-films are

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recognized to be integrable with single-metal gate (midgap) due to their intrinsically low threshold voltage. In this work, we present midgap $CoSi_2$ metal gate by total gate silicidation on SON transistors with Si-conduction channel thickness down to 5 nm. Due to its architecture and to the continuity between SD areas and the bulk, SON transistors allow deep silicidation process down to the gate oxide, meaning that no more polysilicon is left. SON PMOS devices were performed with 55 nm $CoSi_2$ gate length with 5 nm of Si-channel thickness, and show excellent performances (350 μA/μm I_{on} with only 0.1 nA I_{off} @−1.4 V with $T_{ox} = 20$ Å). The poly-depletion is of course suppressed and the gate resistance (<2 Ω/square) is very competitive for RF applications.

2. Why thin film and thin BOX?

The SON stack itself is physically confined to the under-gate-plus-spacer area of a device, thus enabling extremely shallow and highly doped extensions, while leaving the highly doped drain (HDD) junctions comfortably deep. The extension depth is determined by the Si-film thickness, in contrary to bulk devices, that is the key feature permitting the SCE suppression [17]. As presented in Fig. 1 (classical 2D numerical simulation), reducing the conduction film thickness suppress the lateral electrostatic coupling and leads to a SCE and drain induced barrier lowering (DIBL) reduction.

An improved subthreshold slope is often put forward as an argument in favor of SOI devices as far as low-voltage applications are considered. It has been demonstrated that fully depleted SOI with thick buried oxide exhibits extremely low subthreshold slope, close to the ideal value of 60 mV/dec which is definitely lower than the best results measured on bulk devices. This is true for long-channel transistors, but may no longer be valid in the case of short-channel ones.

The superiority of SOI in terms of ideal subthreshold slope is lost in short-channel devices because of the too thick buried oxide that acts as a field guideline. Fig. 2 shows the potential contours for 160 nm thickness of buried oxide and two gate lengths: $L_{gate} = 400$ nm, Fig. 2(a), and $L_{gate} = 80$ nm, Fig. 2(b). In spite of all other geometrical and electrical parameters being strictly identical, the shortest transistor shows a potential distribution typical for DIBL dominated regime of operation. The thick buried oxide reinforces the drain-to-channel coupling, which is out of gate control and thus relaxes the electrostatic integrity of a device.

Consequently, the vertical potential profile near the source is pushed up to positive values (a peak is emerging in the BOX, see Fig. 3) leading also to an increase of the potential in the conduction channel. The potential barrier between the channel and the extensions becomes lower and this leads to the increase in SCE.

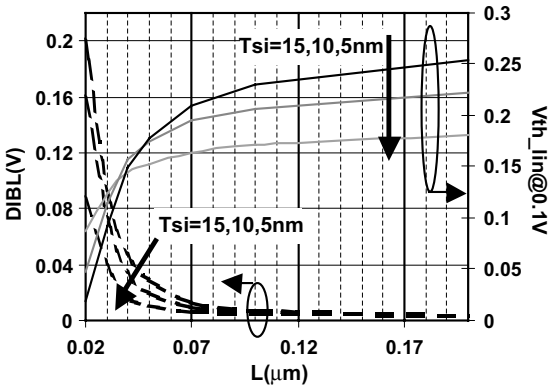


Fig. 1. Reducing the silicon thickness (T_{Si}) improves SCE and DIBL control (2D numerical simulations).

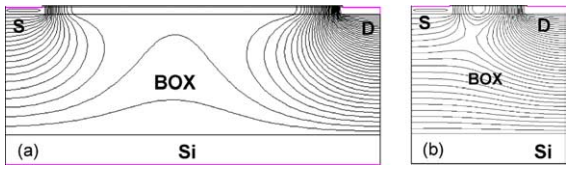


Fig. 2. (a) Equipotential distribution for a 400 nm long device with a 70 nm-thick buried oxide. (b) For shortest gates ($L = 80$ nm), a thick BOX leads to an electrostatic coupling between the source and the drain. (2D numerical simulations).

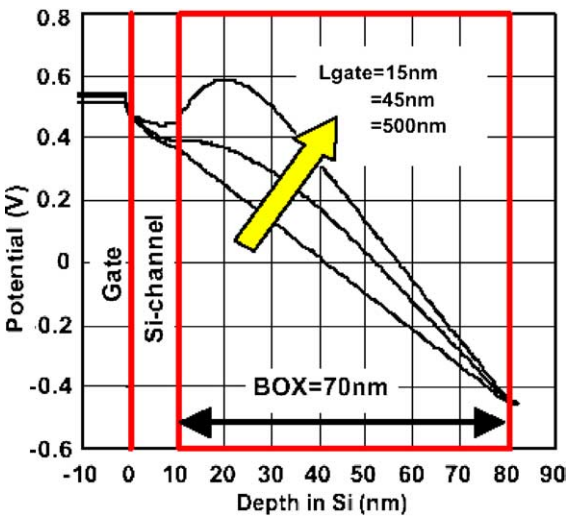


Fig. 3. 1D distribution of the potential in the SON structure. $T_{Si} = 10$ nm, $T_{box} = 70$ nm and $L_{gate} = 500$, 45 and 15 nm. Electrostatic coupling in the BOX leads to a dramatical increase of the potential in the BOX and consequently in the conduction channel. This phenomenon is at the origin of SCE increase.

Decreasing the buried oxide thickness down to 10 nm suppress this lateral electrostatic coupling, so that the potential profile in the structure becomes unperturbed, even for the shortest devices (Fig. 4).

Considering the impact of the substrate doping on the potential distribution, Fig. 5 shows that increasing the bulk doping level leads to a strong reduction of the depletion depth under the BOX and contributes to the reduction of the buried oxide equivalent thickness.

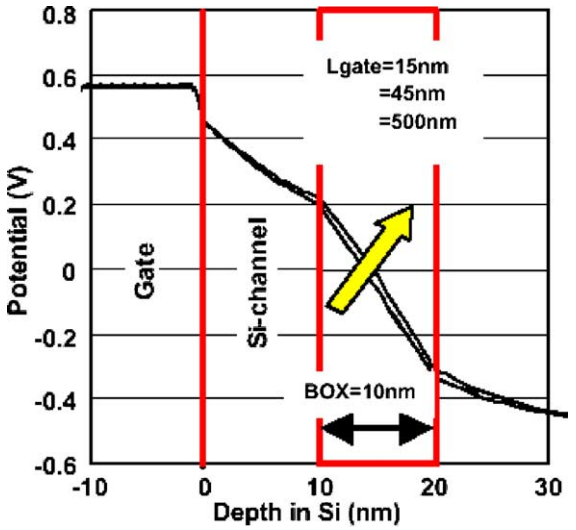


Fig. 4. 1D distribution of the potential in the SON structure. $T_{Si} = 10$ nm, $T_{box} = 10$ nm and $L_{gate} = 500$, 45 and 15 nm. Reducing the BOX thickness suppress the lateral electrostatic coupling so that the device becomes much more immune to parasitic effects.

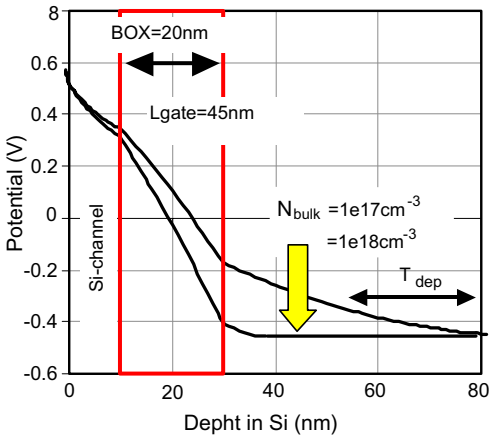


Fig. 5. 1D distribution of the potential in the SON structure. $T_{Si} = 10$ nm, $T_{box} = 20$ nm, $L_{gate} = 45$ nm, $N_{substrate} = 10^{17}$ cm^{-3} and 10^{18} cm^{-3}. When the substrate doping is low, the depletion depth below the BOX increases. This helps the potential increase in the BOX and consequently in the conduction channel.

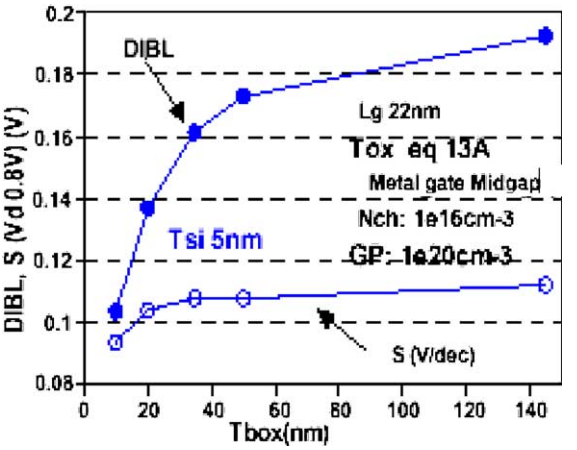


Fig. 6. Simulated FDSOI device with $T_{Si} = 5$ nm, $L_g = 22$ nm $T_{ox_eq} = 13$ Å metal gate, non-doped channel, with ground plane (GP). Curves DIBL and S vs Box thickness. The ultra-thin Si-film (5 nm) and BOX permit to maintain the subthreshold slope below 100 mV/dec and DIBL ~100 mV in the shortest devices, especially for M-G with non-doped channel. Thin Box thickness is efficient for BOX thickness under 40 nm.

Consequently, when lateral electrostatic coupling occurs, the subthreshold slope S increases more rapidly; see Fig. 6. Only SON (or SOI) transistors with very thin silicon films and thin buried oxide can be scaled down to 20 nm channel length still maintaining the subthreshold slope below 100 mV/dec. It is interesting to note that the body factor increases when thinning the buried oxide, leading to an increase of S values for long devices. However, as the devices become more SCE resistive, the subthreshold slope can be kept at low values down to 20 nm gate length.

3. Implementation on SON MOSFETs

As shown in Fig. 7, SON devices were fabricated in the way described in [18–21], where the empty tunnel is fulfilled with the RTO(20 Å)/HTO(50 Å)/Si_3N_4 (300 Å) stack.

As we can see in HR-TEM pictures of the structure (Fig. 7), extremely short devices were fabricated, with gate length down to 38 nm. The Si-conduction channel thickness was measured at 9 nm, without defects in the epi-layer (Fig. 8). In particular, a very important point was to evaluate the formation and morphology of the silicide layer on source and drain junctions.

As shown in the insert in Fig. 7, thanks to the continuity between S/D areas and the bulk, the deposited Co layer reacts with the mono-silicon without any lateral penetration. No voids or cracks, typical when siliciding thin SOI films, are present, leading to a perfect

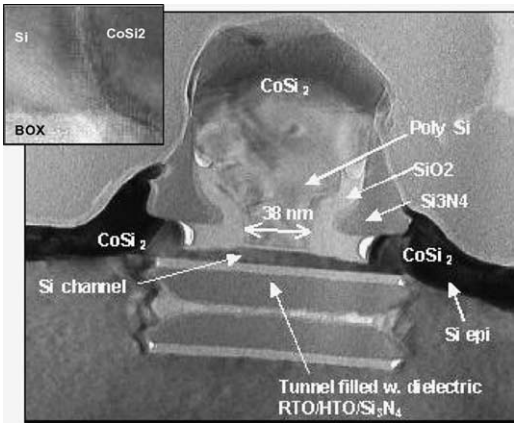


Fig. 7. 38 nm SON PMOS transistor with 9 nm of Si-conduction channel. Junctions are silicided with $CoSi_2$.

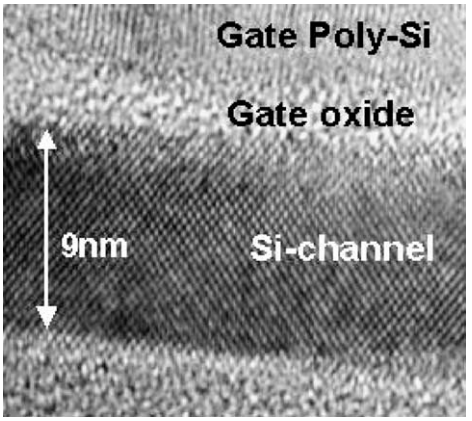


Fig. 8. HR-TEM zoom on the perfect mono-Si layer in the Si-channel. The gate oxide is 20 Å thick.

continuity between the Si-conduction channel and the $CoSi_2$ silicided extensions.

Results are presented on 38 nm PMOS transistors with 20 Å of physical oxide thickness. The best split exhibits drain current as high as 360 μA/μm with 100 nA I_{off} current @$V_{dd} = -1.4$ V (Figs. 9 and 10), with only 60 mV of DIBL for the thinnest conduction channel (9 nm), Fig. 9.

4. Threshold voltage of thin body devices

As the silicon conduction film thickness becomes defined by epitaxy in the SON process, transistors with conduction channel down to 5 nm have been fabricated [16]. Fig. 11 shows this 5 nm-thick Si-channel. Note in particular that top and bottom interfaces are both of similar quality, that demonstrates that the SON tunnel

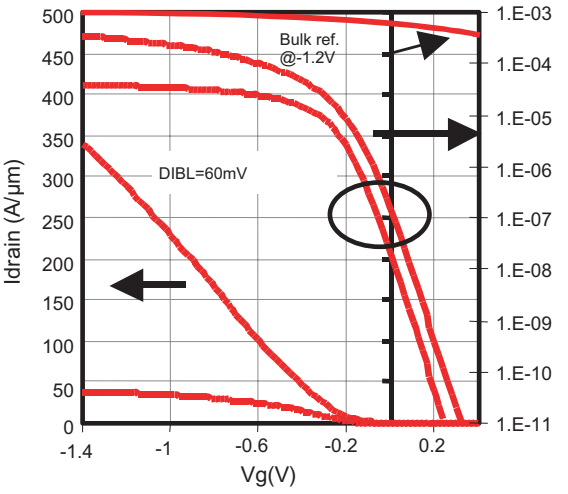


Fig. 9. Measured $\ln I_d(V_g)$ and $\log I_d(V_g)$ of our 38 nm PMOS SON device. The architecture shows high-resistive SCE properties with only 60 mV of DIBL (pockets implanted). I_{off} current is only equal to 100 nA/μm.

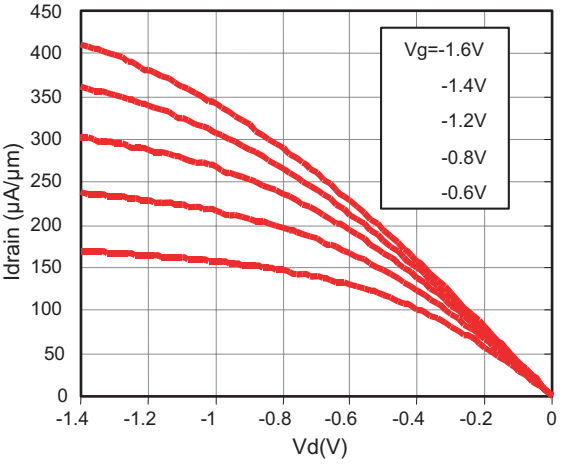


Fig. 10. Measured $I_d(V_d)$ of our 38 nm PMOS SON device. Results show highly performant devices, allowing 360 μA/μm @$V_{dd} = -1.4$ V (consistent with physical $T_{ox} = 20$ Å).

etching is a high-quality and a highly selective process. This point is very important in order to avoid any degradation of the mobility in the channel due to scattering with back interface.

As demonstrated previously, extremely thin film SOI and DG devices are extensively studied for their potential to suppress SCE down to very short channels. However, SCE control for such aggressive gate length as 20 nm needs extremely thin and well-controlled Si-channel thickness down to 5 nm.

A major issue is that V_{th} decreases with the film thickness, leading to intolerably high off current. This is

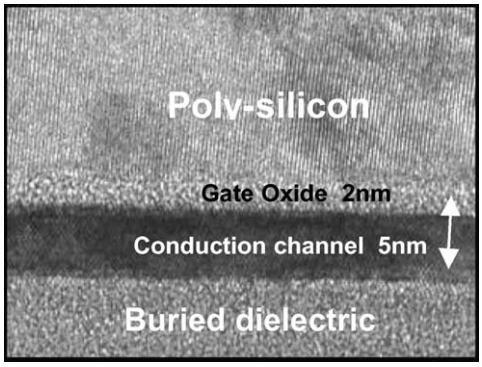


Fig. 11. HR-TEM picture showing the thickness of the Si-channel. Measured $T_{Si} = 5$ nm, with perfect top and bottom interfaces.

due to the depletion charge ($Q_{dep} = q \cdot N_{channel} \cdot T_{Si}$) which is decreasing with T_{Si}. The fabricated devices show excellent DIBL (60 mV) and drive current (350 μA/μm @−1.2 V and 410 μA/μm @−1.4 V with $T_{ox} = 20$ Å, see Fig. 12) for PMOS transistors, but low V_{th} (−0.095 V) leading to 500 nA I_{off}.

The threshold voltage adjustment of FD devices with thin films becomes consequently a serious issue that needs to be addressed.

Fig. 13 represents the evolution of the threshold voltage with increasing level of doping. In order to reach values above 0.2 V for NMOS transistors, the required doping should be above 8×10^{18} cm^{-3} with a 5 nm thin film. All the more, for L_{gate} below 30 nm, this value should be above 10^{19} cm^{-3}. While SCE becomes mainly controlled by the film thickness, the use of such high levels of doping may lead to mobility degradation. This is why metal gates may be mandatory for the end-of-the roadmap devices.

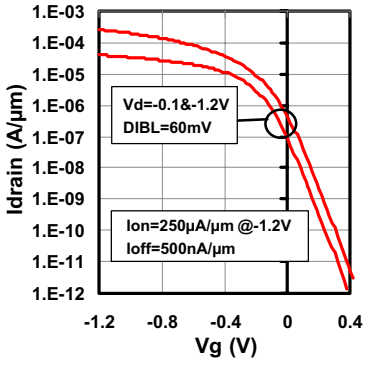


Fig. 12. Measured $\log I_d(V_g)$ of a SON PMOS transistor with $L_{gate} = 55$ nm and $T_{Si} = 5$ nm (implanted pockets). Performances are excellent with only 60 mV of DIBL and a drive current of 350 μA/μm @−1.2 V and 410 μA/μm @−1.4 V. With a polysilicon gate, the corresponding threshold voltage is −0.095 V.

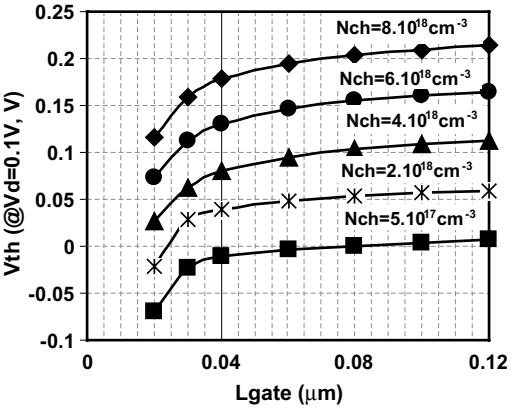


Fig. 13. Influence of the Si-channel doping (N_{ch}) on simulated SON devices. V_{th} adjust needs high channel doping (above $6–8\times10^{18}$ cm^{-3}) than can impact on the mobility of the devices, while SCE are controlled by T_{Si} and T_{box}. Simulation parameters: $T_{Si} = 5$ nm, $T_{ox} = 10$ Å.

5. Integration of SON devices with totally silicided gates

As previously exposed, high doping required for the V_{th} adjustment offsets the intrinsic advantages of FDSOI devices that are capable to suppress SCE with undoped channels. The use of midgap gates allow V_{th} adjustment with lowly doped channels, and may be also useful for polydepletion and gate-resistance reduction. We performed totally silicided gates on SON devices, with very well controlled and reproducible 5 nm thin Si-conduction channel.

The process of the total gate silicidation is the following: (a) the structure is covered with Co+TiN capping and (b) RTP is performed for $CoSi_2$ formation down to the gate oxide. The cobalt silicidation was achieved with two RTA steps (530 °C, 30 s and 830 °C, 30 s) with a cobalt removal after the first anneal. The silicidation was stopped on the gate oxide without any reaction between Co and SiO_2 [22].

5.1. Application to SON devices

SON architecture allows comfortable silicidation thanks to the metallurgical contact between S/D and the bulk. As shown in Fig. 7, the deposited Co layer reacts with the mono-silicon without any lateral penetration. Fig. 14 shows the implementation of the totally silicided gate process to SON devices, with reduced gate height (800 Å) in order to limit silicidation depth in the junctions.

SON transistors were fabricated with this process and Fig. 15 shows that the 800 Å gates were totally transformed into $CoSi_2$.

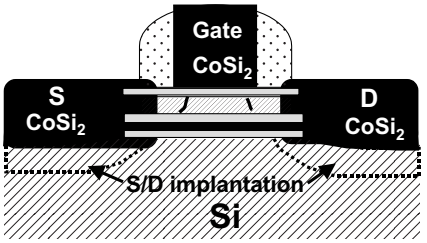


Fig. 14. Implementation of the total gate silicidation process to SON devices. SON architecture allows total gate silicidation process without creating voids in the Si-film. This is thanks to the metallurgical contact between S/D and bulk. Gate height was equal to 800 Å.

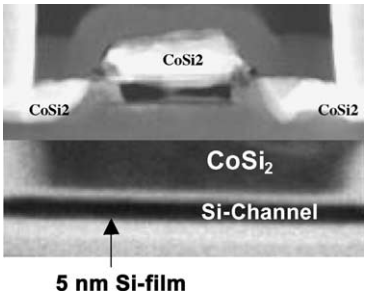


Fig. 15. Example of SON transistor with metal gate integration. Silicidation process induces deep junctions (800 Å depth) that can be included in deliberately deepened SD implantation (double implant, see Fig. 22). Gate height = 800 Å and T_{ox} = 20 Å.

Note that the process of the total gate silicidation without any CMP step on bulk structures leads to unacceptable deep silicided junctions. Therefore on bulk, a CMP step has to be introduced in order to dissociate S/D and gate silicidation that, however, complicates the process a lot. With SON process, the CMP is not needed. This is thanks to the fact that only SON allows fabrication of deep S/D junctions without the danger of a through-the-gate implantation of the S/D dopants into the channel, and without exposing the device to SCE/DIBL. Indeed, the deep S/D junction can be obtained with two low-energy implants (preventing through-the-gate penetration): one after the S/D etching, and the second after S/D epi, Fig. 16.

6. Electrical demonstration

Thanks to the double S/D implantation, I_{diode} leakage of midgap gates SON devices is the same compared to poly-Si gate SON transistors (Fig. 17). $C(V)$ measurements confirm that the $CoSi_2$ gate has suppressed the polydepletion effects (Fig. 18), and the V_{FB} shift is measured to be equal to 0.5 V.

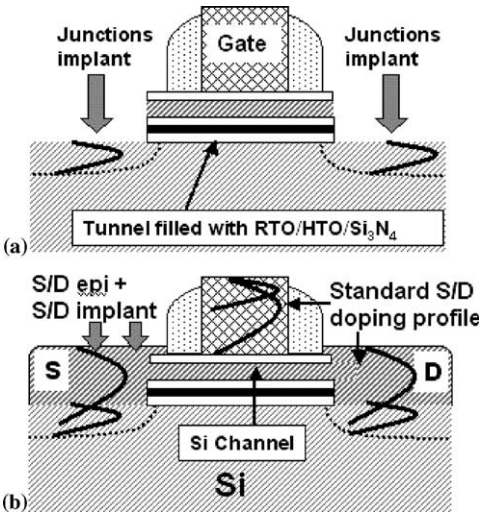


Fig. 16. (a) Shallow implantation after the S/D areas etching. (b) Standard S/D implant after S/D epitaxy. Thank to deeper hard junctions, this process suppresses diode leakage between $CoSi_2$ and the bulk.

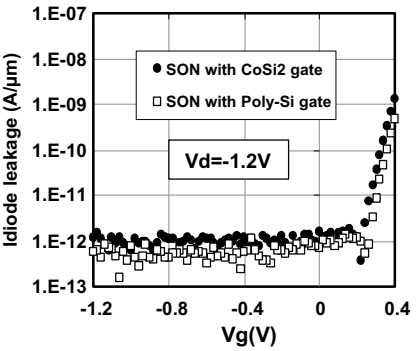


Fig. 17. Measured I_{bulk} leakage of midgap gates SON devices compared to Poly-Si SON transistors. The leakage current is the same, showing that the diode current is not degraded by the deep silicidation process (thanks to double S/D implantation—Fig. 22).

Electrical performances are presented on 55 nm PMOS transistor with 5 nm of conduction channel thickness.

Fig. 19 shows highly SCE-immune devices—only 60 mV of DIBL and a subthreshold slope of 73 mV/dec—in spite of deep S/D junctions. Such results are possible only on SON thanks to its intrinsic strength for SCE/DIBL suppression.

Thank to the V_{th} adjustment with midgap gate ($V_{th} = -0.47$ V @−0.1 V, Fig. 19), I_{off} was reduced to 0.1 nA/µm for 350 µA/µm of drive current @−1.4 V (see Fig. 20).

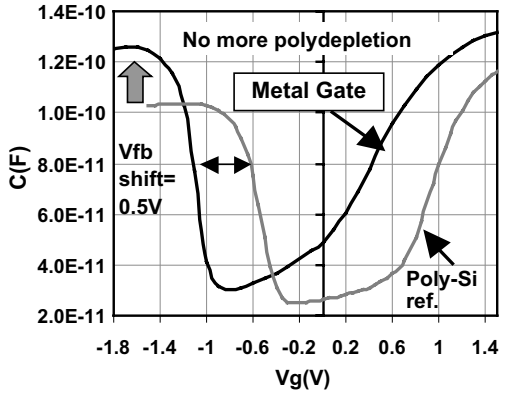


Fig. 18. Measured $C(V)$ curves of PMOS transistors with $W/L = 100$ μm/100 μm. $CoSi_2$ gate suppresses the polydepletion and induces a 0.5 V V_{FB} shift.

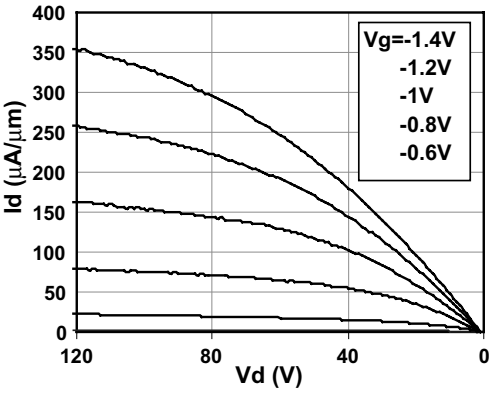


Fig. 20. Measured $\ln I_d(V_d)$ of the 55 nm SON transistor with $CoSi_2$ gate. Drive current as high as 350 μA/μm is measured $@V_g = -1.4$ V. $V_{dd} = -1.4$ V is consistent with $T_{ox} = 20$ Å.

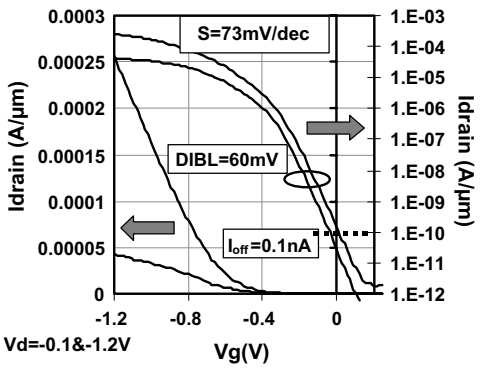


Fig. 19. Measured $\ln I_d(V_g)$ and $\log I_d(V_g)$ of a 55 nm SON transistor with $CoSi_2$ gate. Data show high SCE-resistive devices (DIBL = 60 mV) and thank to the V_{th} shift induced by the midgap gate, I_{off} is around 0.1 nA/μm. The drive current is over 250 μA/μm $@-1.2$ V with $T_{ox} = 20$ Å and over 350 μA/μm $@-1.4$ V.

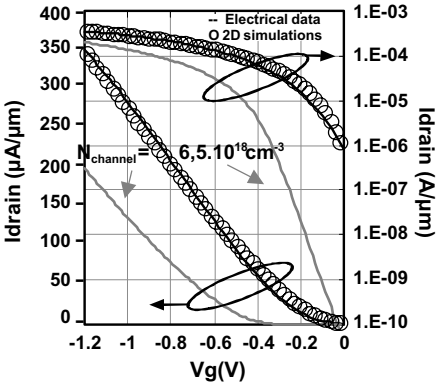


Fig. 21. Calibration of 2D numerical simulations (SON with poly-Si gate, open circles) on our electrical data (black lines). Next the simulations have been extended to the case of poly-gate (gray lines) with adjusted $I_{off} = 0.1$ nA/μm by means of increased doping ($N_{ch} = 6.5 \times 10^{18}$ cm^{-3}). The corresponding $I_{on} = 195$ μA/μm is 25% lower than the 260 μA/μm obtained with $CoSi_2$ gate ($@V_g - V_{th} = -0.8$ V).

If we compare SON devices with poly-Si gate and $CoSi_2$ gates (Fig. 12 vs 19), with the same physical characteristics except the gate material, we can conclude that the use of a midgap gate for V_{th} adjustment on extremely thin Si-film leads to more than four decades of I_{off} reduction, maintaining very good I_{on}. For poly-Si gates, $I_{on} = 210$ μA/μm $@V_g - V_{th} = -0.8$ V and $I_{on} = 280$ μA/μm $@V_g - V_{th} = -1$ V. For $CoSi_2$ gates, $I_{on} = 260$ μA/μm $@V_g - V_{th} = -0.8$ V and $I_{on} = 350$ μA/μm $@V_g - V_{th} = -1$ V.

In fact, the use of the midgap gate is not only to shift the threshold voltage of the device. Thank to the poly-depletion suppression, the equivalent oxide thickness becomes lower and improves SCE control, subthreshold slope as well as the drive current of the device.

Alternative strategy consisting in equivalent I_{off} reduction due to strong channel doping (with poly-gate) leads to very degraded mobility and to a loss of 25% in drive current ($I_{on} = 195$ μA/μm $@V_g - V_{th} = -0.8$ V) as shown by 2D simulations calibrated on the electrical data, Fig. 21.

7. Impact on ITRS specifications

In Fig. 22, we show the impact of the SON architecture on I_{off}/I_{on} specifications. For comparison, the bulk devices with poly-Si gates are also shown, demonstrating large discrepancy with respect to the requirements.

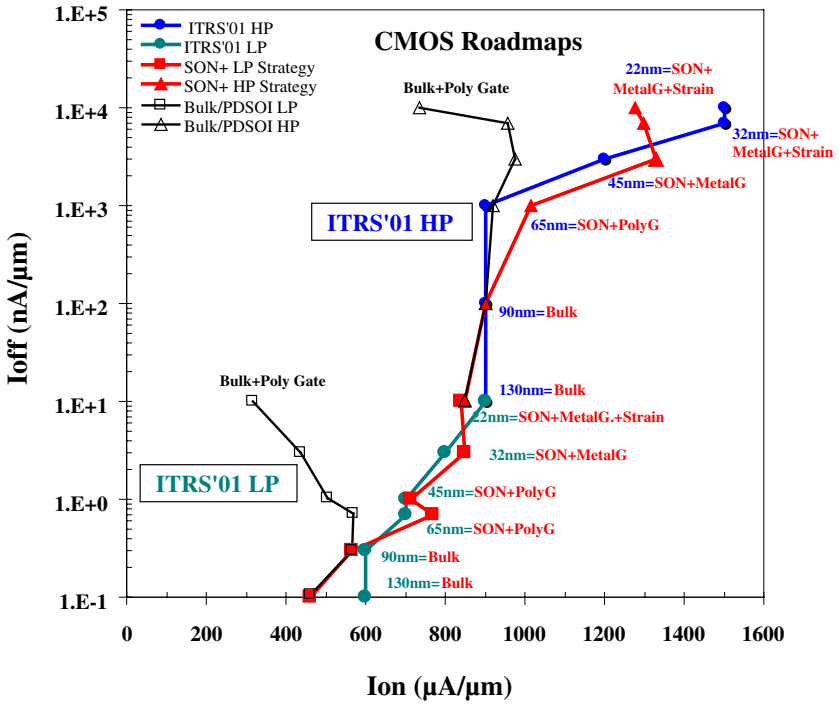


Fig. 22. Impact of SON architecture (analytical simulations) on ITRS specifications.

Results have been performed thanks to analytical simulations based on MASTAR model [23,24].

In contrast, almost the entire LP branch specifications can be reached when replacing bulk with SON. Up to the 45 nm node, SON suffices even with poly-gate. Introduction of metallic gate permits reaching the 32 nm node and approaching closely the 22 nm node. Concerning the HP branch, SON with metal gate enables the 45 nm node, but it is still insufficient for 32 and 22 nm nodes, even with strained channel. There, only double gate (DG) devices have the necessary potential.

8. Conclusions

The SON process allowing fabrication of very thin film SOI-like transistors localized on bulk wafers has been presented. The very thin layers in the SON transistor allow high control of the short-channel effects due to the suppression of any electrostatic lateral coupling (in the channel and in the BOX) and allow in this way excellent electrical performances. The SON architecture gives also additional flexibility to the junction design. Ultra-shallow extensions and sufficiently deep hard junctions also simplify contact and silicidation. SON, as a unique technology, allows a metallic gate by total silicidation without CMP.

It was also demonstrated that SON is better suited than bulk for accepting a metallic gate for low-voltage operation due to the intrinsic low threshold voltage of such thin films. All the more, the SON belongs to the very rare family of processes whose relevant technological window improves when scaling down dimensions. Both etching and filling of the tunnel becomes much easier for short transistors than for longer features. It should be also emphasized that SON is available for all bulk CMOS manufacturers without any special technology changes and investments.

It is also important, that SON devices as originating from bulk CMOS are intrinsically compatible and co-integrable with bulk transistors. As now-a-days technologies have to offer a large versatility of function-dedicated devices on-chip (HP, LP, LL, analog, RF, IO buffer, etc.) this compatibility and co-integrability is considered an important advantage of the SON concept.

Thanks to its merits, SON allows reaching the I_{on}/I_{off} specifications down to the 32 nm node (ITRS'01) and approaching closely those for the 22 nm node, that is by far impossible with bulk. The FDSOI presents similar merits on condition such thin as 5 nm film of Si and BOX can be fabricated.

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